

Midterm 1 Study Guide

Calculus BC • September 2026

Show all work. Use the blank space after each question. A complete answer key and source list begin after the practice set.

- 1.** Use an algebraic simplification to help find the limit, if it exists.

$$\lim_{x \rightarrow 25} \frac{\sqrt{x} - 5}{x - 25}.$$

- 2.** For $f(x) = \frac{x+5}{|x+5|}$ and $a = -5$, find each limit, if it exists:

$$(a) \lim_{x \rightarrow a^-} f(x) \quad (b) \lim_{x \rightarrow a^+} f(x) \quad (c) \lim_{x \rightarrow a} f(x).$$

- 3.** If $f(x) = 2x^2 + 1$, then $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x^2}$ is

- (A) 0
- (B) 1
- (C) 2
- (D) 4
- (E) nonexistent

4. For the given limit $\lim_{x \rightarrow 4} x^2 = 16$ and $\varepsilon = 0.1$, use the graph of f to find the largest δ such that if $0 < |x - 4| < \delta$, then $|f(x) - 16| < \varepsilon$.

5. Use the ε - δ definition of a limit to prove that

$$\lim_{x \rightarrow 2} (5x - 3) = 7.$$

6. What is $\lim_{h \rightarrow 0} \frac{8(\frac{1}{2} + h)^8 - 8(\frac{1}{2})^8}{h}$?

- (A) 0
- (B) $\frac{1}{2}$
- (C) 1
- (D) The limit does not exist.
- (E) It cannot be determined from the information given.

7. Use theorems on limits to find the limit, if it exists.

$$\lim_{x \rightarrow -3} \frac{x + 3}{(1/x) + (1/3)}.$$

8. If $0 \leq f(x) \leq c$ for some real number c , prove that $\lim_{x \rightarrow 0} x^2 f(x) = 0$.

9. $\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$ is

- (A) 0
- (B) 1
- (C) $\sin x$
- (D) $\cos x$
- (E) nonexistent

10. For $f(x) = \frac{8}{(2x+5)^3}$ and $a = -\frac{5}{2}$, express each limit as ∞ , $-\infty$, or DNE (Does Not Exist):

- (a) $\lim_{x \rightarrow a^-} f(x)$ (b) $\lim_{x \rightarrow a^+} f(x)$ (c) $\lim_{x \rightarrow a} f(x)$.

11. Find the vertical and horizontal asymptotes for the graph of f .

$$f(x) = \frac{x^2 - 5x}{x^2 - 25}.$$

12. If k is a positive integer, then $\lim_{x \rightarrow +\infty} \frac{x^k}{e^x}$ is

- (A) 0
- (B) 1
- (C) e
- (D) $k!$
- (E) nonexistent

13. Classify the discontinuities of f as removable, jump, or infinite.

$$f(x) = \begin{cases} -x^2, & x < 1, \\ 2, & x = 1, \\ x - 2, & x > 1. \end{cases}$$

14. If $f(x) = x^3 - 5x^2 + 7x - 9$, use the intermediate value theorem to prove that there is a real number a such that $f(a) = 100$.

- 15.** Let f be defined by

$$f(x) = \begin{cases} \sin x, & x < 0, \\ x^2, & 0 \leq x < 1, \\ 2 - x, & 1 \leq x < 2, \\ x - 3, & x \geq 2. \end{cases}$$

For what values of x is f NOT continuous?

- (A) 0 only
- (B) 1 only
- (C) 2 only
- (D) 0 and 2 only
- (E) 0, 1, and 2

- 16.** Let $f(x) = 3 - 2x^2$. (a) Use the limit definition to find the slope of the tangent line to the graph of f at $P(a, f(a))$. (b) Find an equation of the tangent line at $P(2, f(2))$.

- 17.** A projectile is fired directly upward from the ground with an initial velocity of 112 ft/sec. Its distance above the ground after t seconds is $112t - 16t^2$ feet. (a) Find the velocity of the projectile at $t = 2$, $t = 3$, and $t = 4$. (b) When does the projectile strike the ground? (c) Find the velocity at the instant it strikes the ground.

- 18.** For what nonnegative value of b is the line $y = -\frac{1}{3}x + b$ normal to the curve $y = x^3$?

- (A) 0
- (B) 1
- (C) $\frac{4}{3}$
- (D) $\frac{10}{3}$
- (E) $\frac{10\sqrt{3}}{3}$

- 19.** Let $f(x) = x^3 - 4x$ and $P(2, 0)$. (a) Use the definition of the derivative to find $f'(x)$. (b) Find the domain of f' . (c) Find an equation of the tangent line to the graph of f at P . (d) Find the points on the graph at which the tangent line is horizontal.

- 20.** Is $f(x) = \sqrt[3]{x}$ differentiable on the given interval? Explain. (a) $[-1, 1]$; (b) $[-2, -1]$.

- 21.** If f is a function such that $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = 0$, which statement must be true?

- (A) The limit of $f(x)$ as x approaches 2 does not exist.
- (B) f is not defined at $x = 2$.
- (C) $f'(2) = 0$.
- (D) f is continuous at $x = 0$.
- (E) $f(2) = 0$.

22. Find the derivative of $G(v) = \frac{v^3 - 1}{v^3 + 1}$.

23. Solve the equation $\frac{d^2y}{dx^2} = 0$ for $y = 6x^4 + 24x^3 - 540x^2 + 7$.

24. If u , v , and w are nonzero differentiable functions, then the derivative of $\frac{uv}{w}$ is

(A) $\frac{uv' + u'v}{w'}$

(B) $\frac{u'v'w - uvw'}{w^2}$

(C) $\frac{uvw' - u'vw - uv'w}{w^2}$

(D) $\frac{u'vw + uv'w + uvw'}{w^2}$

(E) $\frac{uv'w + u'vw - uvw'}{w^2}$

25. Find the derivative of $h(z) = \frac{1 - \cos z}{1 + \cos z}$.

26. Find equations of the tangent line and the normal line to the graph of $f(x) = \csc x + \cot x$ at the point $(\pi/4, f(\pi/4))$.

27. If $f(x) = \frac{x}{\tan x}$, then $f'(\frac{\pi}{4}) =$

- (A) 2
- (B) $\frac{1}{2}$
- (C) $1 + \frac{\pi}{2}$
- (D) $\frac{\pi}{2} - 1$
- (E) $1 - \frac{\pi}{2}$

- 28.** Let $y = 1/x^2$, $a = 3$, and $\Delta x = 0.3$. (a) Find general formulas for Δy and dy . (b) If, for the given values of a and Δx , x changes from a to $a + \Delta x$, find the values of Δy and dy .
- 29.** Constriction of arterioles is a cause of high blood pressure. It has been verified experimentally that, as blood flows through an arteriole of fixed length, the pressure difference between the two ends of the arteriole is inversely proportional to the fourth power of the radius. If the radius of an arteriole decreases by 10%, use differentials to find the percentage change in the pressure difference.
- 30.** Let $f(x) = x^2 - 2x + 3$. The tangent line to the graph of f at $x = 2$ is used to approximate $f(x)$. Which is the greatest value of x for which the error is less than 0.5?
- (A) 2.4
 - (B) 2.5
 - (C) 2.6
 - (D) 2.7
 - (E) 2.8

- 31.** Find the derivative of $f(x) = \sin \sqrt{x} + \sqrt{\sin x}$.
- 32.** Let p , q , and r be functions such that $p(z) = q(r(z))$. If $r(3) = 3$, $q(3) = -2$, $r'(3) = 4$, and $q'(3) = 6$, find $p(3)$ and $p'(3)$.
- 33.** Let f and g be differentiable, with $f(1) = 2$, $f'(1) = 3$, $f'(2) = -4$, $g(1) = 2$, $g'(1) = -3$, and $g'(2) = 5$. If $h(x) = f(g(x))$, then $h'(1) =$
- (A) -9
 - (B) -4
 - (C) 0
 - (D) 12
 - (E) 15
- 34.** Assuming that the equation $x = \sin(xy)$ determines a differentiable function f such that $y = f(x)$, find y' .

- 35.** Assuming that the equation $3x^2 + 4y^2 = 4$ determines a function f such that $y = f(x)$, find y'' , if it exists.
- 36.** The slope of the line tangent to the curve $y^2 + (xy + 1)^3 = 0$ at $(2, -1)$ is
- (A) $-\frac{3}{2}$
 - (B) $-\frac{3}{4}$
 - (C) 0
 - (D) $\frac{3}{4}$
 - (E) $\frac{3}{2}$
- 37.** A metal rod has the shape of a right circular cylinder. As it is being heated, its length is increasing at a rate of 0.005 cm/min and its diameter is increasing at 0.002 cm/min. At what rate is the volume changing when the rod has length 40 centimeters and diameter 3 centimeters?

- 38.** A ladder 20 feet long leans against a vertical building. If the bottom of the ladder slides away from the building horizontally at a rate of 2 ft/sec, at what rate is the angle between the ladder and the ground changing when the top of the ladder is 12 feet above the ground?

- 39.** A person 2 meters tall walks directly away from a streetlight 8 meters above the ground. The person's shadow is lengthening at $\frac{4}{9}$ meter per second. At what rate, in meters per second, is the person walking?

- (A) $\frac{4}{27}$
- (B) $\frac{4}{9}$
- (C) $\frac{3}{4}$
- (D) $\frac{4}{3}$
- (E) $\frac{16}{9}$

- 40.** Suppose f and g are functions such that $f(2) = -1$, $f'(2) = 4$, $f''(2) = -2$, $g(2) = -3$, $g'(2) = 2$, and $g''(2) = 1$. Find the value of each of the following at $x = 2$: (a) $(2f - 3g)'$; (b) $(2f - 3g)''$; (c) $(fg)'$; (d) $(fg)''$; (e) $\left(\frac{f}{g}\right)'$; (f) $\left(\frac{f}{g}\right)''$.

- 41.** Let $f(x) = \begin{cases} (2x - 1)^3, & x \geq 2, \\ 5x^2 + 34x - 61, & x < 2. \end{cases}$ Determine if f is differentiable at 2.

- 42.** If f and g are twice differentiable and $h(x) = f(g(x))$, then $h''(x) =$

- (A) $f''(g(x))[g'(x)]^2 + f'(g(x))g''(x)$
- (B) $f''(g(x))g'(x) + f'(g(x))g''(x)$
- (C) $f''(g(x))[g'(x)]^2$
- (D) $f''(g(x))g''(x)$
- (E) $f''(g(x))$

- 43.** Find the extrema of f on the given interval.

$$f(x) = 3x^2 - 10x + 7, \quad [-1, 3].$$

- 44.** Find the critical numbers of $k(r) = r^5 - 2r^3 + r - 12$.

- 45.** If $f(x) = \frac{1}{3}x^3 - 4x^2 + 12x - 5$ on $0 \leq x \leq 9$, then the absolute maximum value of f occurs when x is

- (A) 0
- (B) 2
- (C) 4
- (D) 6
- (E) 9

- 46.** Show that f satisfies the hypotheses of Rolle's theorem on $[a, b]$, and find all c in (a, b) such that $f'(c) = 0$.

$$f(x) = x^4 + 4x^2 + 1, \quad [-3, 3].$$

- 47.** Determine whether f satisfies the hypotheses of the mean value theorem on $[a, b]$ and, if so, find all c in (a, b) such that $f(b) - f(a) = f'(c)(b - a)$.

$$f(x) = x^3 - 2x^2 + x + 3, \quad [-1, 1].$$

- 48.** If $f'(x)$ and $g'(x)$ exist and $f'(x) > g'(x)$ for all real x , then the graphs of $y = f(x)$ and $y = g(x)$
- (A) intersect exactly once.
 - (B) intersect no more than once.
 - (C) do not intersect.
 - (D) could intersect more than once.
 - (E) have a common tangent at each intersection.

- 49.** Find the local extrema of f , the intervals on which f is increasing or decreasing, and sketch its graph.

$$f(x) = 10x^3(x - 1)^2.$$

- 50.** Find the local extrema of f on $[0, 2\pi]$, the subintervals on which f is increasing or decreasing, and sketch its graph.

$$f(x) = \frac{1}{2}x - \sin x.$$

- 51.** Sketch a continuous function f satisfying all the conditions:

$$\begin{aligned}f(0) &= 3, & f(-2) &= f(2) = -4, & f'(0) &\text{ is undefined,} \\f'(-2) &= f'(2) = 0, \\f'(x) &> 0 \text{ if } -2 < x < 0 \text{ or } x > 2, \\f'(x) &< 0 \text{ if } x < -2 \text{ or } 0 < x < 2.\end{aligned}$$

- 52.** Find the local extrema of f , using the second derivative test whenever applicable. Find the intervals of concavity, the x -coordinates of the inflection points, and sketch the graph.

$$f(x) = \sqrt[5]{x} - 1.$$

- 53.** Find the local extrema of f , using the second derivative test whenever applicable. Find the intervals of concavity, the x -coordinates of the inflection points, and sketch the graph.

$$f(x) = \sqrt[3]{x^2}(3x + 10).$$

- 54.** Sketch a continuous function f satisfying all the conditions:

$$\begin{aligned} f(0) &= 1, & f(2) &= 3, & f'(0) &= f'(2) = 0, \\ f'(x) &< 0 \text{ if } |x - 1| > 1, & f'(x) &> 0 \text{ if } |x - 1| < 1, \\ f''(x) &> 0 \text{ if } x < 1, & f''(x) &< 0 \text{ if } x > 1. \end{aligned}$$

55. Sketch a continuous function f satisfying all the conditions:

$$\begin{aligned}f(0) &= 2, & f(2) &= f(-2) = 1, & f'(0) &= 0, \\f'(x) &> 0 \text{ if } x < 0, & f'(x) &< 0 \text{ if } x > 0, \\f''(x) &< 0 \text{ if } |x| < 2, & f''(x) &> 0 \text{ if } |x| > 2.\end{aligned}$$

56. Sketch a continuous function f satisfying all the conditions:

$$f(1) = 4, \quad f'(x) > 0 \text{ if } x < 1, \quad f'(x) < 0 \text{ if } x > 1, \quad f''(x) > 0 \text{ for } x \neq 1.$$

- 57.** Sketch a continuous function f satisfying all the conditions. If n is odd, $f(n) = 1$ and $f'(n) = 0$; if n is even, $f(n) = 0$ and $f'(n)$ does not exist. For every integer n ,

$$f'(x) > 0 \quad \text{when } 2n < x < 2n + 1,$$

$$f'(x) < 0 \quad \text{when } 2n - 1 < x < 2n,$$

$$f''(x) < 0 \quad \text{when } 2n < x < 2n + 2.$$

Answer Key

Answers correspond to questions 1–57.

1. $\frac{1}{10}$.
2. (a) -1 ; (b) 1 ; (c) DNE.
3. (C) 2 .
4. $\delta = \sqrt{16.1} - 4 \approx 0.0124805$.
5. For any $\varepsilon > 0$, choose $\delta = \varepsilon/5$. If $0 < |x - 2| < \delta$, then $|(5x - 3) - 7| = 5|x - 2| < 5\delta = \varepsilon$.
6. (B) $\frac{1}{2}$.
7. -9 .
8. $0 \leq x^2 f(x) \leq cx^2$, and both bounding functions tend to 0 as $x \rightarrow 0$. The sandwich theorem gives $\lim_{x \rightarrow 0} x^2 f(x) = 0$.
9. (D) $\cos x$.
10. (a) $-\infty$; (b) ∞ ; (c) DNE.
11. Vertical: $x = -5$. Horizontal: $y = 1$.
12. (A) 0 .
13. Removable discontinuity at $x = 1$; $\lim_{x \rightarrow 1} f(x) = -1 \neq f(1) = 2$.
14. f is continuous on $[6, 7]$, and $f(6) = 69 < 100 < 138 = f(7)$. By the intermediate value theorem, some $a \in (6, 7)$ satisfies $f(a) = 100$.
15. (C) 2 only.
16. (a) $m = -4a$. (b) $y = -8x + 11$.
17. (a) 48 , 16 , and -16 ft/sec, respectively. (b) $t = 7$ sec. (c) -112 ft/sec.
18. (C) $\frac{4}{3}$.
19. (a) $f'(x) = 3x^2 - 4$. (b) $(-\infty, \infty)$. (c) $y = 8x - 16$. (d) $\left(-\frac{2\sqrt{3}}{3}, \frac{16\sqrt{3}}{9}\right)$ and $\left(\frac{2\sqrt{3}}{3}, -\frac{16\sqrt{3}}{9}\right)$.
20. (a) No: $f'(0)$ does not exist as a finite number. (b) Yes: $f'(x) = 1/(3x^{2/3})$ exists throughout the interval (with one-sided derivatives at its endpoints).
21. (C) $f'(2) = 0$.
22. $G'(v) = \frac{6v^2}{(v^3 + 1)^2}$.
23. $x = -5$ or $x = 3$.
24. (E) $\frac{uv'w + u'vw - uvw'}{w^2}$.

25. $h'(z) = \frac{2 \sin z}{(1 + \cos z)^2}$.

26. Tangent: $y - (1 + \sqrt{2}) = -(2 + \sqrt{2})(x - \pi/4)$. Normal: $y - (1 + \sqrt{2}) = \frac{x - \pi/4}{2 + \sqrt{2}}$.

27. (E) $1 - \frac{\pi}{2}$.

28. (a) $\Delta y = \frac{1}{(x + \Delta x)^2} - \frac{1}{x^2}$ and $dy = -\frac{2}{x^3} dx$, where $dx = \Delta x$. (b) $\Delta y = -\frac{7}{363}$ and $dy = -\frac{1}{45}$.

29. The pressure difference increases by approximately 40%.

30. (D) 2.7.

31. $f'(x) = \frac{\cos \sqrt{x}}{2\sqrt{x}} + \frac{\cos x}{2\sqrt{\sin x}}$ (where the terms are defined).

32. $p(3) = -2$ and $p'(3) = 24$.

33. (D) 12.

34. $y' = \frac{1 - y \cos(xy)}{x \cos(xy)}$.

35. $y'' = -\frac{3}{4y^3}$ for $y \neq 0$.

36. (D) $\frac{3}{4}$.

37. The volume is increasing at $\frac{21\pi}{160}$ cm³/min.

38. $\frac{d\theta}{dt} = -\frac{1}{6}$ rad/sec; the angle is decreasing.

39. (D) $\frac{4}{3}$ m/s.

40. (a) 2; (b) -7; (c) -14; (d) 21; (e) $-\frac{10}{9}$; (f) $-\frac{19}{27}$.

41. Yes. Both pieces have value 27 at $x = 2$, and both one-sided derivatives equal 54, so $f'(2) = 54$.

42. (A) $f''(g(x))[g'(x)]^2 + f'(g(x))g''(x)$.

43. Absolute maximum 20 at $x = -1$; absolute minimum $-\frac{4}{3}$ at $x = \frac{5}{3}$.

44. $r = -1, -\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 1$.

45. (E) 9.

46. The polynomial is continuous and differentiable, and $f(-3) = f(3) = 118$. The only number is $c = 0$.

47. Yes. The average slope is 2, and $c = \frac{2-\sqrt{7}}{3}$ is the only solution in $(-1, 1)$.

48. (B) intersect no more than once.

49. Increasing on $(-\infty, \frac{3}{5}]$ and $[1, \infty)$; decreasing on $[\frac{3}{5}, 1]$. Local maximum $f(\frac{3}{5}) = \frac{216}{625}$; local minimum $f(1) = 0$. The critical number $x = 0$ is not an extremum.
50. Decreasing on $[0, \frac{\pi}{3}]$ and $[\frac{5\pi}{3}, 2\pi]$; increasing on $[\frac{\pi}{3}, \frac{5\pi}{3}]$. Local minimum $f(\frac{\pi}{3}) = \frac{\pi}{6} - \frac{\sqrt{3}}{2}$; local maximum $f(\frac{5\pi}{3}) = \frac{5\pi}{6} + \frac{\sqrt{3}}{2}$.
51. One valid answer is $f(x) = \frac{7}{4}(|x| - 2)^2 - 4$. It has local minima at $(-2, -4)$ and $(2, -4)$ and a cusp-shaped local maximum at $(0, 3)$.
52. No local extrema; increasing on $(-\infty, \infty)$. Concave up on $(-\infty, 0)$ and down on $(0, \infty)$. Inflection point $(0, -1)$, with a vertical tangent.
53. Local maximum $f(-\frac{4}{3}) = 4\sqrt[3]{6}$; local minimum $f(0) = 0$. Concave down on $(-\infty, 0)$ and $(0, \frac{2}{3})$; concave up on $(\frac{2}{3}, \infty)$. Inflection point $(\frac{2}{3}, 4\sqrt[3]{12})$; cusp at the origin.
54. Decreasing on $(-\infty, 0)$, increasing on $(0, 2)$, and decreasing on $(2, \infty)$. Local minimum $(0, 1)$; local maximum $(2, 3)$. Concave up for $x < 1$, concave down for $x > 1$; inflection at $x = 1$ (height not determined).
55. Increasing on $(-\infty, 0)$ and decreasing on $(0, \infty)$, with local maximum $(0, 2)$. Concave up on $(-\infty, -2)$ and $(2, \infty)$; concave down on $(-2, 2)$. Inflection points $(-2, 1)$ and $(2, 1)$.
56. Increasing on $(-\infty, 1)$ and decreasing on $(1, \infty)$, with local maximum $(1, 4)$. Concave up on both sides of $x = 1$. One valid shape has a cusp at $(1, 4)$, for example $f(x) = 4 - \sqrt{|x - 1|}$.
57. One valid graph is $f(x) = |\sin \frac{\pi x}{2}|$: concave-down arches rise from each even-integer corner $(2n, 0)$ to a horizontal maximum $(2n + 1, 1)$ and return to $(2n + 2, 0)$.

Where These Questions Came From

This guide covers textbook Sections 2.1 through 4.4. For 2.1–4.2, each group uses two textbook exercises not listed anywhere in the 2026–27 calendar’s Solve, Recommended Continued Practice, Opener, Challenge, or IP columns, plus one question from the supplied AP packet. The final nine questions reproduce every 4.3–4.4 textbook exercise already collected in the 4.3–4.6 study guide.

Section 2.1

1 = 2.1 #18

2 = 2.1 #26

3 = AP Packet p. 13, BC 2 (1993)

Section 2.2

4 = 2.2 #9

5 = 2.2 #16

6 = AP Packet p. 4, BC 6 (1969)

Section 2.3

7 = 2.3 #28

8 = 2.3 #65

9 = AP Packet p. 5, BC 10 (1988)

Section 2.4

10 = 2.4 #3

11 = 2.4 #34

12 = AP Packet p. 2, BC 35 (1988)

Section 2.5

13 = 2.5 #16

14 = 2.5 #59

15 = AP Packet p. 2, BC 5 (1988)

Section 3.1

16 = 3.1 #2

17 = 3.1 #16

18 = AP Packet p. 3, BC 4 (1973)

Section 3.2

19 = 3.2 #4

20 = 3.2 #22

21 = AP Packet p. 3, BC 8 (1985)

Section 3.3

22 = 3.3 #19

23 = 3.3 #45

24 = AP Packet p. 4, BC 4 (1988)

Section 3.4

25 = 3.4 #13

26 = 3.4 #30

27 = AP Packet p. 5, BC 6 (1985)

Section 3.5

28 = 3.5 #3

29 = 3.5 #41

30 = AP Packet p. 4, BC 92 (1998)

Section 3.6

31 = 3.6 #53

32 = 3.6 #82

33 = AP Packet p. 7, BC 39 (1973)

Section 3.7

34 = 3.7 #12

35 = 3.7 #29

36 = AP Packet p. 8, BC 3 (1998)

Section 3.8

37 = 3.8 #34

38 = 3.8 #42

39 = AP Packet p. 9, BC 37 (1988)

Section 3.9

40 = 3.9 #63

41 = 3.9 #66

42 = AP Packet p. 7, BC 5 (1998)

Section 4.1

43 = 4.1 #8

44 = 4.1 #16

45 = AP Packet p. 13, BC 27 (1973)

Section 4.2

46 = 4.2 #5

47 = 4.2 #17

48 = AP Packet p. 3, BC 15 (1969)

Section 4.3

$$49 = 4.3 \#7$$

$$50 = 4.3 \#19$$

$$51 = 4.3 \#35$$

Section 4.4

$$52 = 4.4 \#9$$

$$53 = 4.4 \#11$$

$$54 = 4.4 \#31$$

$$55 = 4.4 \#33$$

$$56 = 4.4 \#34$$

$$57 = 4.4 \#37$$